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CSE 350: Digital Electronics and Pulse techniques

Exp-05: Flash Analog to Digital converter (ADC)

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Objectives

1. To analyze a 2-bit flash analog to digital converter.

Equipment and component list

Equipment

1. Digital Multimeter
2. Trainer board

Component

- Single Supply Quad Operational Amplifier - LM324 - x1 piece
- 8-to-3 Line Priority Encoder - IC74148 - x1 piece
- Resistors -
  - 10 KΩ - x4 pieces

Task-01: Flash ADC

THEORY

Flash ADC is the fastest analog-to-digital converter. You can see the circuit diagram of a 2-bit flash ADC in figure 1. All the op-amps operate as comparators in this circuit. The analog input (V_A) is applied to the 'inverting' input of the three op-amps.

There is a resistive ladder-network with a reference voltage V_{REF} = 5 V at the top of the network. We will obtain some fixed voltages at each node of this network. These nodes are denoted as V_1, V_2 and V_3 . Then, we connected the V_1 node to op-amp 1 (OA1). Similarly, the other two nodes are connected to the corresponding op-amps.

Now, let us calculate the node voltages V_i's of the ladder network. For this, keep in mind that the current towards the op-amp's input terminals are negligible. First, the total resistance of the ladder network is

$$R_{Total} = \Sigma R_i = R_1 + R_2 + R_3 + R_4 \qquad (1)$$

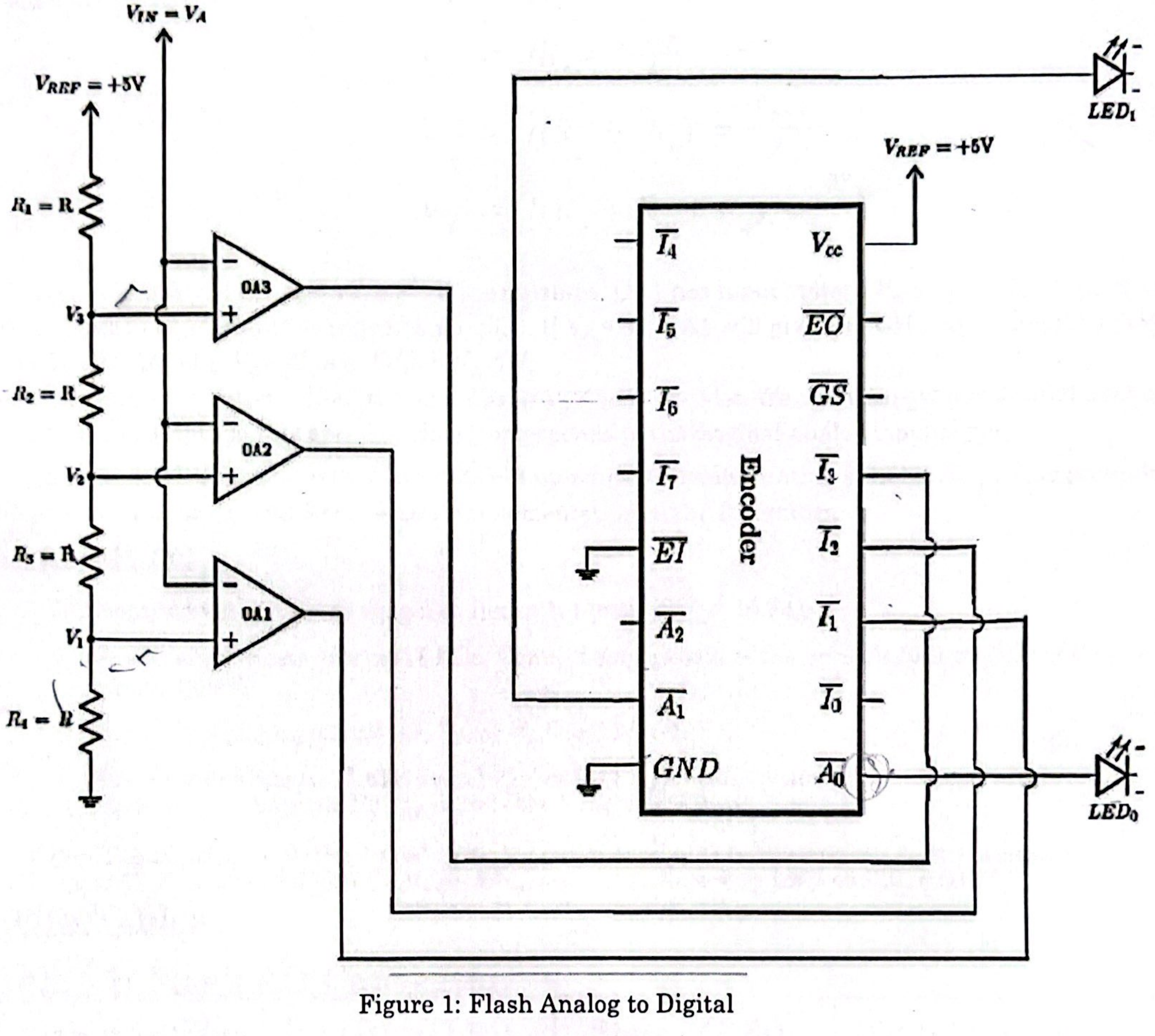
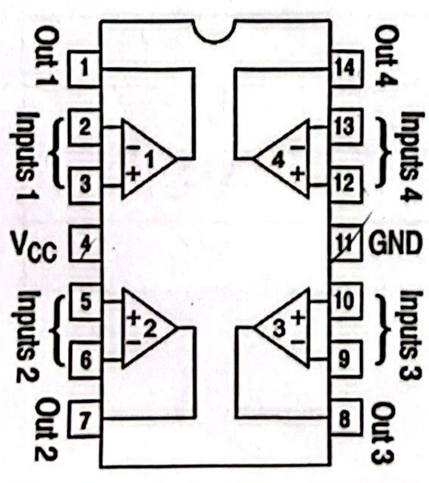
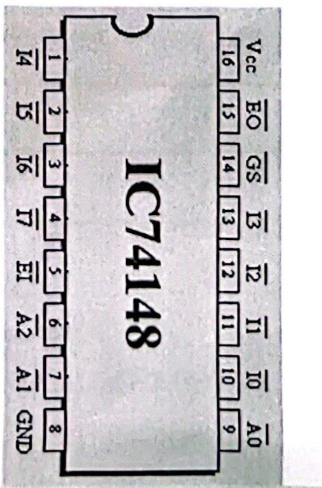


Figure 1: Flash Analog to Digital



LM324 IC (Quad Op-Amp) pin diagram



74148 IC (Encoder) pin diagram

So, using Ohm's law, the current through the ladder network will be (same current flows through all the Ri's

$$I_{ladder} = \frac{V_{REF} - 0}{R_{Total}} = \frac{V_{REF}}{4R} \qquad (2)$$

It is now trivial to calculate all the node voltages. The equations for all the node voltages are given below for your convenience.

$$V_1 = IR_4 = \frac{V_{REF}}{4} \qquad (3)$$

$$V_2 = I(R_3 + R_4) = \frac{V_{REF}}{2} \qquad (4)$$

$$V_3 = I(R_2 + R_3 + R_4) = \frac{3V_{REF}}{4} \qquad (5)$$

Now, closely analyze the operation of all the op-amps. OA1 has input voltage V_A at its '–' input (inverting input) and V_1 at '–' input (non-inverting input). If $V_A < V_1$, OA1 will give a HIGH output. Similarly, OA2 will give HIGH output if $V_A < V_2$ and OA3 if $V_A < V_3$.

Next, we send the outputs of all the op-amps to a priority encoder. We will then get our desired 2-bit digital signal at the output of this encoder which corresponds to the original analog input signal.

For this flash ADC design, we will need $2^n - 1$ op-amps for implementing an n-bit ADC. This presents a huge disadvantage in terms of practical implementation in the laboratory.

Procedure:

1. Construct the circuit as shown in figure 1. Consider, $R = 10\ K\Omega$.
2. We will not use any external LEDs. Connect the outputs of the encoder to the LEDs of the Trainer Board.
3. Vary the analog input voltage, V_{IN} or V_A from 0V to 5V.
4. Observe when the two LEDs switch ON or OFF and measure the input voltage which causes the transitions. Fill up data table 1 using this data.

Note: The encoder is "Active LOW". This means that whenever the output (A_0, A_1) is supposed to be "Logical 1", they are at a LOW voltage. Hence, the corresponding LED will turn OFF!

Data Tables

Table 1: Flash AD Converter.

Input Voltage $V_{IN} = V_A$	State of LED_1	State of LED_0	Digital Binary Output
0V	ON	ON	00
1.27V	ON	OFF	01
2.6V	OFF	ON	10
3.8V	OFF	OFF	11

Table 1: Data Table for Flash AD Converter

Lab Task 1

Use your "group number" as input voltage V_A and observe the output. If the group number is greater than 5, divide it by 2 and use the resultant value as input. Explain the reason for obtaining the output.

Our group no is 10. $\therefore V = \frac{10}{2} = 5V$
so. $V_A = V_{in} = 5V$. we can see that our both led are off.
so the digital output is 11. $A_0 = 280.2mV$

$A_1 = 201.1 mV$.

Lab Task 2

Adjust the input voltage such that we get Binary output 00 and 01. For each case, measure the output voltages of the encoder. Explain why the LEDs turn on or off. (<u>Note: disconnect the LEDs when measuring the output voltages</u>)

Output bits. A_1A_0	Output voltage (V)	
00	5.03V	4.6 V
01	2.4V	0.508V

Lab Task 3

Measure the voltages of points V_3, V_2 and V_1. Do the values match with the theory?

Using VDR, $V_1 = \frac{10}{4\times10} \times 5 = 1.25V$

$V_2 = \frac{2\times10}{4\times10} \times 5 = 2.5 V$

$V_3 = \frac{3\times10}{4\times10} \times 5 = 3.75V$

so the values are close to theory.

V_1 (V)	V_2 (V)	V_3 (V)
1.24V	2.49V	3.78V

Signature

Report

1. Write down an advantage and a disadvantage of a Flash AD converter.

Ans.

Advantage: This converter is very fast as it requires one clock cycle to convert the analog to digital data

Disadvantage: It requires a lot components (Resistors, op-amps, encoders), as a result, the power consumption is high.

2. If we wanted to build a 3-bit Flash AD converter, how many resistors and comparators (op-amps) would we need?

Ans.

For a 3-bit Flash AD converter,

$n = 3$

Resistors needed $= 2^n = 2^3 = 8$

and

Comparators needed $= 2^n - 1 = 2^3 - 1$

$= 8 - 1 = 7$

3. Write a discussion and include the following: your overall experience, accuracy of the measured data, difficulties experienced, and your thoughts on those.

This experiment helped me understand the working of a 2-bit flash ADC, especially how comparators and encoders convert analog signals into digital ~~signals~~ outputs. The overall experience connected theory with practical application.

The measured output data was mostly accurate, with minor deviations due to component tolerances. One mistake that we did was, not powering up the quad-op-amp IC by connecting pin 4 to Vcc and 11 to GND initially. So, our circuit was incomplete and we later fixed it. We also had an issue with a loose wire causing incorrect output, which we fixed after inspecting the connections. After that, the output became stable and it was as expected. Also, the variable voltage knob of the trainer board gave us a really hard time while calculating values for task-2.

To sum up, this experiment improved my circuit troubleshooting skills and will be helpful for future labs involving digital and mixed-signal systems.